



Nazmi Yücel Çan

MACHINE LEARNING | DATA SCIENTIST

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PROFESSIONAL SUMMARY

I am an Information Systems and Technologies graduate and a Machine Learning Engineer / Data Scientist focused on deploying and optimizing AI models in production environments. I co-authored an academic paper with my team members, published in the EI-Cezeri Journal of Science and Engineering (2026), on deep learning models integrating attention mechanisms (Attention U-Net/U-Net++) for military camouflaged object detection. During my internship at Deep Think Technology, I developed industrial computer vision systems featuring real-time YOLOv8-seg inference, PyQt5 GUI design, and camera SDK integration. I am experienced in optimizing Small Language Models (SLMs like mT5) and Edge AI/TinyML applications (SVM on Arduino) using ONNX Runtime and INT8 quantization for resource-constrained devices. I ranked in the top 10 finalists of the Pupilica AI Hackathon with the practical solutions I developed. My primary focus lies in end-to-end model development workflows—from data preprocessing and model training to deployment and optimization in computer vision and natural language processing.

SKILLS

Programming Languages

Python (Advanced), SQL, Bash/Shell Scripting

ML & AI Frameworks

TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face (LoRA/PEFT), ONNX Runtime

Core Domains

Computer Vision (OpenCV, YOLOv8), NLP (Transformers), Data Preprocessing, Model Evaluation & Quantization

Tools & Platforms

Git/GitHub, Linux (Ubuntu/Debian), PyQt5, Google Gemini API, VS Code, Jupyter Notebooks

Languages: Turkish (Native), English (Intermediate)

EXPERIENCE

Machine Learning Intern at Deep Think Technology

02/2026 – 06/2026

- Developed an "Apple Detection and Quality Analysis System" using YOLOv8-seg for real-time fruit detection and CIEDE2000 color quality analysis.
- Designed and implemented a desktop GUI with PyQt5, integrating camera SDK, ML inference, and physical dimension measurement modules.
- Trained and deployed computer vision models on embedded devices for real-time edge AI inference.
- Gained hands-on experience in data preprocessing, model evaluation, and the practical challenges of edge computing.

PROJECTS

Arduino PC System Monitor

Python, Arduino (C++), PlatformIO, TinyML (SVM), scikit-learn

- A TinyML-powered system that monitors PC hardware resources (CPU, RAM, GPU, VRAM) and automatically changes the power mode by classifying the current load using an SVM model running on an Arduino Leonardo.

GitHub

Apple Detection and Quality Analysis System

Python, PyQt5, OpenCV, YOLOv8, MVS SDK

- An industrial desktop application built with PyQt5 for real-time apple detection, physical dimension measurement, and CIEDE2000 color quality analysis using YOLOv8-seg. Developed during an internship, I focused on designing the GUI and integrating the core system modules.

GitHub

mT5 Bidirectional EN-TR Machine Translation (ONNX INT8 & ORT Mobile)

Python, PyTorch, Hugging Face PEFT (LoRA), ONNX Runtime, INT8 Quantization, SLM

- Fine-tuned an mT5-small model with 8-bit LoRA for bidirectional English-Turkish translation (EN<->TR). The model was optimized using ONNX and INT8 quantization specifically for mobile deployment via ONNX Runtime (ORT Mobile). Achieved a METEOR score of 0.6245 while significantly reducing VRAM usage.

Kaggle

Zettelkasten AI Notes

Python, PyQt5, SQLite, Google Gemini AI

- I built this desktop application with PyQt5 to make knowledge management more efficient using the Zettelkasten method. It allows you to create, edit, and link notes with full Markdown support and real-time previews, storing everything locally in an SQLite database. I also integrated Google Gemini AI to automatically generate Zettelkasten-style notes from PDF documents and suggest relevant links between your notes.

GitHub

PUBLICATIONS

Deep Learning Models Integrating Attention Mechanisms For Military Camouflaged Object Detection — Nilgün Şengöz, Gül Karaman, Mert Samet Çeliker, Nazmi Yücel Çan
El-Cezeri Fen ve Mühendislik Dergisi (2026) — <https://doi.org/10.31202/ecjse.1747013>

EDUCATION

Bachelor of Science in Information Systems and Technologies 09/2022 – 06/2026
Burdur Mehmet Akif Ersoy University
GPA: 3.17/4.00

REFERENCES

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